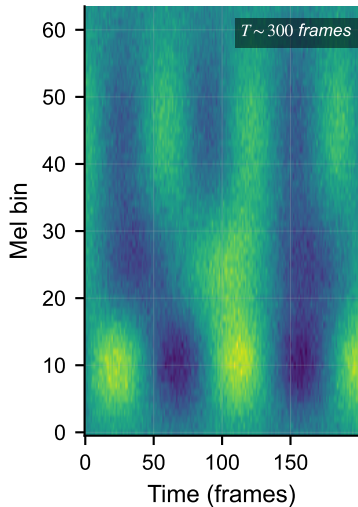
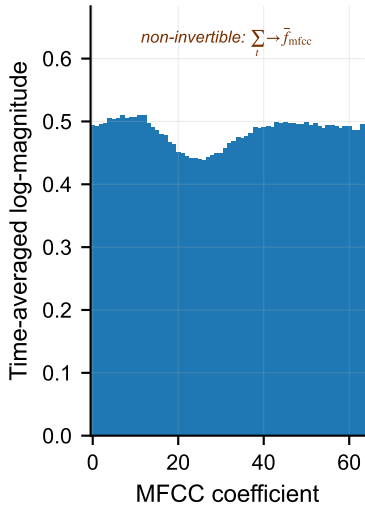


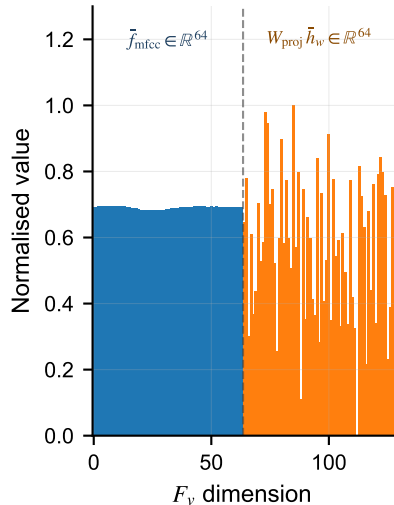
(a) Original log-mel spectrogram



(b) Time-averaged 64-d MFCC



(c) Final 128-d F_v for transmission



Both steps (time-averaging + global pooling) drop $O(T)$ bits of frame-level information; WER ≥ 0.95 under white-box + black-box GLO attacks